# T-105 ENGINEERING THERMODYNAMIC

## UNIT 2

1. **State the first law of thermodynamics. (NOV, MAY 2017, JAN 2016,)**

The first law of thermodynamics states that the general principle of the conservation of energy. Energy is neither created nor destroyed, but only gets transformed from one form to another.

During a cycle, a system undergoes, the cyclic integral of heat added is equal to the cyclic integral of work done, mathematically, $(\sum W)_{cycle} = (\sum Q)_{cycle}$ or $\oint dW = \oint dQ$

First law of thermodynamics also states that the heat transfer equal to sum of the work done and change in internal energy $Q = W + \Delta U$.

2. **What are the limitations of first law of thermodynamics? (NOV, 2016, MAY 2019)**

- First law of thermodynamics does not give any information on whether thatchange of state or the process is at all feasible or not.
- The first law cannot indicate whether a metallic bar of uniform temperature can spontaneously become warmer at one end and cooler at the other.
- According to the first law of thermodynamics heat and work are mutually convertible during any cycle of closed system. But in actual practice, heat does not convert completely into work.

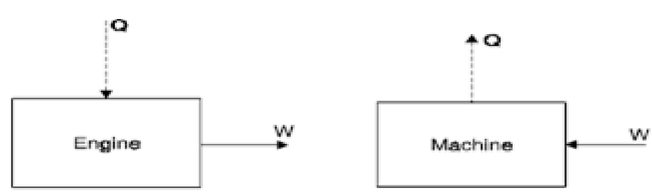
3. **State the characteristic features of an adiabatic process. (NOV 2017)**

An adiabatic process is a thermodynamic process, in which there is no **heat** transfer into or out of the system ($Q = 0$). The system can be considered to be perfectly insulated. In an adiabatic process, **energy** is transferred only as work.

4. **What is perpetual motion machine of first kind? (NOV 2016, JAN 2016)**

There can be no machine which would continuously supply mechanical work without some other form of energy disappearing simultaneously such a fictitious machine is called a perpetual motion machine of the first kind, or PMM1. A PMM1 is thus impossible.

The converse of the above statement is also true, there can be no machine which would continuously consume work without some other form of energy appearing simultaneously.




PPM 1.

The converse of PMM 1.

5. **Give the expression for work done in constant pressure process and constant volume process. (MAY 2016)**

Workdone for constant pressure Process

$$W = P_1(V_2 - V_1)$$

(or)

$$W = mR(T_2 - T_1)$$

Workdone for constant volume Process

$$W = 0$$

Where
$P_1$ = initial pressure (bar)
$T_2$ = final temperature (K)
$T_1$ = initial temperature (K)
$V_2$ = final Volume ($m^3$)
$V_1$ = initial Volume ($m^3$)
$R$ = characteristic gas constant $\cdot$ KJ/kgK
$m$ = mass of substance. (kg)

6. **Write the SFEE for flow process. (MAY 2016)**

$$m\,(gz_1 + C_1^2/2 + u_1 + p_1v_1) + Q = m\,(gz_2 + C_2^2/2 + u_2 + p_2v_2) + W$$

Where m = mass of the substance.

$Z_1$ & $Z_2$ = Height above the datum levels for inlet and outlet in ‘m’.

$u_1$ & $u_2$ = Specific internal energy of working substance entering and exit of the system.J/kg.

$p_1$ & $p_2$ = pressures of working substance entering and exit of the system.(bar)

$v_1$ & $v_2$ = Specific volume of working substance entering and exit of the system ($m^3$/kg).

$C_1$ & $C_2$ = velocity of working substance entering and exit of the system (m/s).

Q = Heat supplied to the system in J/kg.

W = Work delivered by the system in J/kg.

7. **What is meant by steady flow system? (JAN 2015)**

The steady state of a system any thermodynamic property will have a fixed value at a particular location and will not alter with time

The steady flow means that the states of flow of mass and energy across the control surface are constant.

8. **What is the formula for heat transfer equivalent in polytropic process?(JAN 2015)**

Heat transfer for polytropic process $Q = W\left[\frac{\gamma - n}{\gamma - 1}\right]$

Where

$W$ = workdone (kJ)

$\gamma$ = adiabatic index

$n$ = polytropic index

$Q$ = Heat transfer (kJ)

9. **State the first law of thermodynamics for a nonflow process and for a cycle. (NOV 2018)**

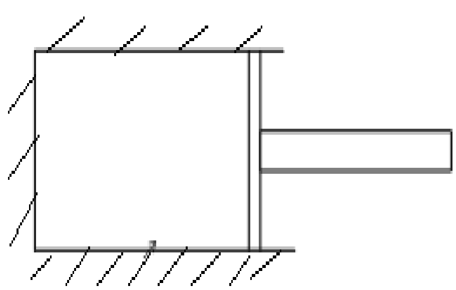
During a cycle, a system undergoes, the cyclic integral of heat added is equal to the cyclic integral of work done, mathematically,

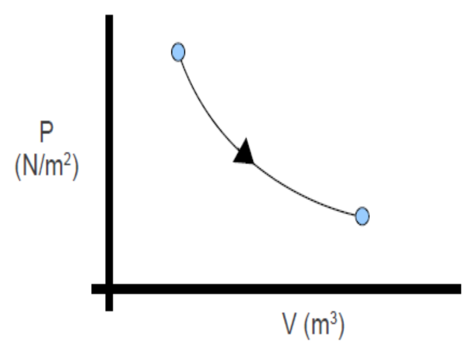
$(\sum W)_{cycle} = (\sum Q)_{cycle}$ or $\oint dW = \oint dQ$

Where the symbol $\oint$ denotes the cyclic integral for the closed path. This is the first law for a non-flow process undergoing a cycle.

10. **What is meant by adiabatic process?(MAY 2017)**

- During adiabatic process, the gas neither receives nor rejects heat. i.e. no heat transfer takes place during adiabatic process.
- Imagine a gas present inside a cylinder made of insulated walls. The gas in the cylinder may be compressed or expanded without heat transfer.





11. **Give the expression for work done in a polytropic process. (MAY 2018)**

Workdone for polytropic process

$$W = \frac{P_1V_1 - P_2V_2}{n-1} \quad \text{(or)}$$

$$= \frac{mR(T_1 - T_2)}{n-1}$$

Where

$P_1, P_2$ = initial & final pressure (bar)

$V_1, V_2$ = initial & final volume ($m^3$)

$n$ = Polytropic Index

$R$ = characteristic gas constant (kJ/kg k)

$m$ = mass of the substance. (kg)

12. **State the first law of thermodynamics applied to flow process. (MAY 2018)**

It states that the rate of heat added Q to the control volume plus the rate at which energy is flowing into the control volume as a result of mass rate of flow $m_{in}$ is equal to rate of change of stored in the control volume $(dE/dt)_{cv}$ plus the rate at which energy is flowing out of the control volume as result of mass rate of flow $m_{out}$ and the rate of work output W

$$Q + m_{in}\left[h + \frac{c^2}{2} + gz\right]_{in} = \left(\frac{dE}{d\tau}\right)_{cv} + m_{out}\left[h + \frac{c^2}{2} + gz\right]_{out} + W$$

13. **Show internal energy is a property of the system (NOV 2018) (Refer First unit questipn no.2 from part b )**

14. **Assume that a gas is heated at a constant volume process. Write the expression for work done and change in internal energy. (SEP 2020)**

Constant volume process

$$\boxed{\text{Workdone} = 0}$$

$\Delta U$ = change in internal Energy

$$\boxed{\Delta U = m c_v (T_2 - T_1)} \text{ KJ}$$

$m$ = mass of the gas (kg)

$c_v$ = Specific heat at constant volume $\frac{KJ}{kg\,K}$.

$T_2$ = final temperature (K)

$T_1$ = Initial temperature (K)

15. **A closed system receives an input heat of 400 kJ and increases the internal energy of the systems for 320 kJ. By applying the first law of thermodynamics. Determine the work done by the system. (SEP 2020)**

given data

$Q = 400 \text{ KJ}$

$\Delta U = 320 \text{ KJ}$

To find

$W = ?$

Sol

By First Law of T.D

$Q = W + \Delta U$

$400 = W + 320$

$W = 400 - 320$

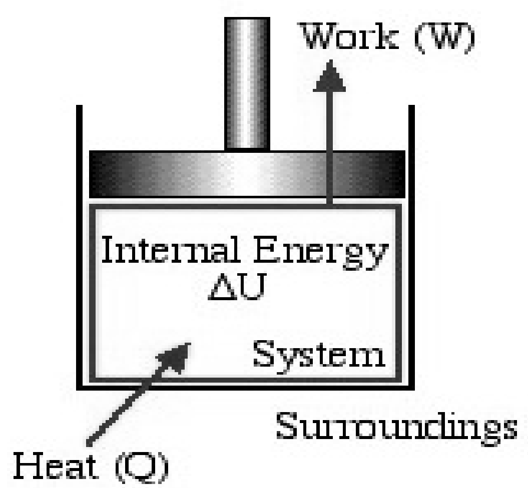
$\boxed{W = 80 \text{ KJ}}$

**11 Marks**

1. **Write a critical note on the conservation of energy principle for closed and open system. (NOV 2017)**

**Conservation of energy law for closed system**

We consider the First Law of Thermodynamics applied to stationary closed systems as a conservation of energy principle. Thus energy is transferred between the system and the surroundings in the form of heat and work, resulting in a change of internal energy of the system. Internal energy change can be considered as a measure of molecular activity associated with change of phase or temperature of the system and the energy equation is represented as follows:




Energy Equation for Stationary Closed Systems

$$\mathbf{Q - W = \Delta U \quad in\ kJ}$$

Dividing each term by the system mass m [kg] we obtain the specific form of the Energy Equation:

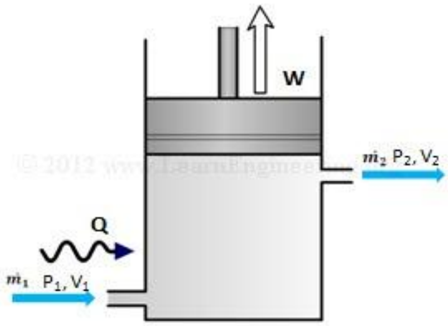
$q – w = \Delta u$ **in kJ/kg**

**Conservation of energy law for open system**

It states that the rate of heat added Q to the control volume plus the rate at which energy is flowing into the control volume as a result of mass rate of flow $m_{in}$ is equal to rate of change of stored in the control volume $(dE/dt)_{cv}$ plus the rate at which energy is flowing out of the control volume as result of mass rate of flow $m_{out}$ and the rate of work output W

$$\Sigma m_{in} = \Sigma m_{out}$$

If the state and flow-rate of the streams leaving and entering the system also does not change with time, it is called steady state system and according to the principle of conservation of energy, the First Law, the rate at which energy is entering the system, will be equal to the rate at which energy flows out of the system. The rate at which energy flows is given in kJ/secor kW.




We have encountered several forms of energy up to this point: enthalpy, work, heat, potential and kinetic energy. Consider an adiabatic system where work is performed and the changes in kinetic and potential energy can be ignored.

$$Q + \dot{m}_{in}\left[h + \frac{c^2}{2} + gz\right]_{in} = \left(\frac{dE}{d\tau}\right)_{cv} + m_{out}\left[h + \frac{c^2}{2} + gz\right]_{out} + W$$

2. **Three kilo grams of nitrogen gas at 6 atm and 160°C is expanded twice the initial volume adiabatically in a cylinder fitted with frictionless piston. It is then compressed to initial volume by constant pressure and then compressed to its initial state by constant volume process. Draw the P-V diagram and calculate the net work done. (MAY 2016)**

given data

$m = 3\,kg$

$P_1 = 6\,atm = 608\ kN/m^2$

$T_1 = 160^\circ c = 433\,K$

$V_2 = 2V_1$

$V_2/V_1 = 2$

1-2 Adiabatic process

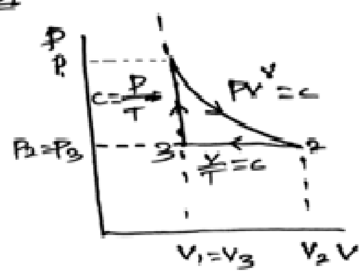
2-3 Constant pressure

3-1 Constant volume

To Find

* Draw the pv diagram
* net work done

Sol




1-2 Adiabatic process.

Characteristic gas equation

$P_1V_1 = mRT_1$ $\qquad$ $R = 0.287\ kJ/kgK.$

$608 \times V_1 = 3\times 0.287 \times 433$

$V_1 = 0.613\,m^3$

$V_2 = 2V_1$

$V_2 = 1.226\,m^3$

Pressure and temperature for adiabatic process

$$\frac{P_1}{P_2} = \left(\frac{V_2}{V_1}\right)^{\gamma}$$

$$P_2 = \frac{608}{\left(\frac{1.226}{0.613}\right)^{1.4}} \qquad \gamma = 1.4$$

$$\underline{\underline{P_2 = 230.3 \text{ kn/m}^2}}$$

$$W_{1-2} = \frac{P_1V_1 - P_2V_2}{\gamma - 1}$$

$$= \frac{608 \times 0.613 - 230 \times 1.226}{1.4 - 1}$$

$$\boxed{W_{12} = 226.81 \text{ KJ}}$$

2 - 3 process Constant pressure process

$$W = P_2(V_3 - V_2)$$

$V_1 = V_3$ Constant volume process

$$W = P_2(V_1 - V_2)$$

$$= 230.3 \times (0.613 - 1.226)$$

$$= \boxed{W = -141.48 \text{ KJ}}$$

3 - 1 process Constant volume process

$\boxed{W_{3-1} = 0 \text{ KJ}}$ for Constant Volume

total net workdone

$$dW = W_{1-2} + W_{2-3} + W_{3-1}$$

$$= 226.81 - 141.48 + 0$$

$$\boxed{dW = 84.88 \text{ KJ}}$$

3. **A gas contained in a cylinder is compressed from *1* MPa and 0.05 $m^3$ to 2 MPa. Compression is governed by $pv^{1.4}$ constant. Internal energy of gas is given by U = (7.5 PV-425) kJ. Where P is pressure in kPa and V is volume in $m^3$. Determine heat, work and change in internal energy assuming compression process to be quasi static. Also find out work interaction, if the 180 kJ of heat is transferred to system between same states. Also explain why it is different from above? (JAN 2016)**

Given data

$P_1 = 1\,MPa = 1000\,kn/m^2$

$V_1 = 0.5\,m^3$

$P_2 = 2\,MPa = 2000\,kn/m^2$

$PV^{1.4} = C$ (Adiabatic process)

$\gamma = 1.4$

$U = 7.5\,PV - 425$

$Q = 180\,KJ$

To find

a) $Q, W, \Delta U$

b) W if heat of 180 KJ transferred to system between same states.

c) Explain why it is different from above

Sol a)

$U = 7.5\,PV - 425$

$\Delta U = U_2 - U_1$

$U_2 = 7.5\,P_2V_2 - 425, \quad U_1 = 7.5\,P_1V_1 - 425$

$\Delta U = 7.5\,(P_2V_2 - P_1V_1)$

Pressure & temperature relation for adiabatic process

$$\frac{P_1}{P_2} = \left(\frac{V_2}{V_1}\right)^{\gamma}$$

$$\frac{V_2}{V_1} = \left(\frac{P_1}{P_2}\right)^{1/\gamma} \qquad \gamma = 1.4.$$

$$V_2 = 0.5\left(\frac{1000}{2000}\right)^{1/1.4}$$

$$\underline{\underline{V_2 = 0.3045\,m^3}}$$

$\Delta U = 7.5\,(2000 \times 0.304 - 1000 \times 0.5)$

$\boxed{\Delta U = 810 \text{ KJ}}$

for Quasi process.

$$W = \frac{P_1V_1 - P_2V_2}{\gamma - 1}$$

$$= \frac{1000 \times 0.5 - 2000 \times 0.304}{1.4 - 1}$$

$\boxed{W = -270 \text{ KJ}}$

$Q = W + \Delta U$

$= -270 + 810$

$\boxed{Q = 540 \text{ KJ}}$

b) Since the End states are the same. ΔU would remain the same

$W = Q - \Delta U$

$W = 180 - 810$

$\boxed{W = -630 \text{ KJ}}$

the work in b is not equat to workdone a since the process is not Quasi static.

4. **The properties of a system, during a reversible constant pressure non-flow process at P = 1.6 bar, changed from $v_1$= 0.3$m^3$/kg. $T_1$ = 20 °C to $v_2$ = 0.55 $m^3$/kg $T_2$ = 260°C. The Specific heat of the fluid isgiven by $C_P$ = 1.5 + (75/(T+45)) kJ/kg°C. Where T is in °C. Determine the heat added, work done , change in internal energy and change in enthalpy per kg of fluid. (JAN 2016)**

Given data ;

$P = C = 1.6$ bar

$V_1 = 0.3\ m^3/kg$

$T_1 = 20°C \longrightarrow 293K$

$V_2 = 0.55\ m^3/kg$

$T_2 = 260°C \longrightarrow 533K$

$C_p = 1.5 + \left(\frac{75}{T_1 + 45}\right)\ KJ/kg\ °C$

To find,

$Q, W, \Delta U, \Delta H$

Solution,

$W = P(V_2 - V_1)$

$= 160 \times (0.55 - 0.3)$

$\boxed{W = 40\ KJ/kg}$

$C_p = ?$

$Q = mC_p\, dt$

$dQ = C_p\, dt$

$\int_1^2 dQ = \int_1^2 C_p . dT$

$$= \int_1^2 \left(1.5 + \frac{75}{T+45}\right) dT$$

$$Q_2 - Q_1 \text{ (or) } Q_{12} \text{ (or) } (Q) = \int_1^2 (1.5)\, dT + \int_1^2 \left(\frac{75}{T+45}\right) dT$$

$$= 1.5 \int_1^2 dT + 75 \int_1^2 \frac{1}{T+45}\, dT$$

$$= 1.5 \left[T\right]_1^2 + 75 \left[\ln\left[T+45\right]\right]_1^2$$

$$= 1.5\left[T_2 - T_1\right] + 75\left[\ln (T_2+45) - \ln (T_1+45)\right]$$

$$= 1.5\left[260-20\right] + 75\left[\ln (260+45) - \ln (20+45)\right]$$

$$= 1.5\left[240\right] + 75\left[5.720 - 4.174\right]$$

$$= 360 + 75\left[1.546\right]$$

$$\boxed{Q = 475.94 \text{ KJ/kg}}$$

$$Q = \Delta H$$

$$\boxed{\Delta H = 475.94 \text{ KJ/kg}}$$

$$Q = W + \Delta U$$

$$\Delta U = Q - W$$

$$= 475.94 - 40$$

$$\boxed{\Delta U = 435.94 \text{ KJ/kg}}$$

5. **One kg of air is compressed slowly in a cylinder and piston arrangement from 1 bar, 200 K to 6 bar. During the process heat is exchanged with the surroundings so that the temperature remains constant. Determine (a) change in internal energy (b) work interaction (c) heat transferred. (MAY 2017)**

given data

$m = 1\ kg$

$P_1 = 1\ bar = 100\ kN/m^2$

$T_1 = 200\ K$

$P_2 = 6\ bar = 600\ kN/m^2$

Constant temperature process $PV = C$

To find

$W,\ \Delta U,\ Q$

Sol

constant volume process.

characteristic gas equ.

$P_1 V_1 = m R T_1$

$100 \times V_1 = 1 \times 0.287 \times 200 \qquad R = 0.287\ kJ/kgK.$

$V_1 = 0.574\ m^3$

P.V relation for constant temperature.

$\frac{P_1}{P_2} = \frac{V_2}{V_1}$

$V_2 = \frac{P_1}{P_2} \times V_1$

$= \frac{100}{600} \times 0.574$

$V_2 = 0.095\ m^3$

a) Workdone (W) = $P_1 V_1 \ln(V_2/V_1)$

$= 1 \times 100 \times 0.574 \times \ln\left(\frac{0.095}{0.574}\right)$

$\boxed{W = -103.24\ KJ}$

b) Change in Internal Energy $\Delta U$

$\boxed{\Delta U = 0}$ for constant temperature process

c) heat transferred

$Q = W + \Delta U$ (First law of thermodynamics)

$\Delta U = 0$

$Q = W$

$\boxed{Q = -103.24\ kJ}$

6. **1 kg of gas at 1.1 bar and 27°C is expanded to 6.6 bar according to the law $pv^{1.3} = C$ calculate work, heat transfer if**
   a. **The gas is ethane with the molecular weight of 30 mol and $C_p$ = 2.1kJ/kg K**
   b. **The gas is argon with molecular weight of 40 mol and $C_p$ = 0.52 kJ/kgK (JAN 2015)**

given data

$m = 1\,kg$

$P_1 = 1.1\,bar = 110\,kN/m^2$

$T_1 = 27°C = 300K$

$P_2 = 6.6\,bar = 660\,kN/m^2$

$pv^{1.3} = c$

Mw for ethane = 30 mol

Mw for Argon = 40 mol

Cp for ethane = 2.1 KJ/kg K

Cp for Argon = 0.52 kJ/kgK.

To find: Q, W for both gas:-

Sol

a) Ethane gas:-

$$W = \frac{mR(T_1 - T_2)}{n-1}$$

T & P relation for polytropic process

$$\frac{T_1}{T_2} = \left(\frac{P_1}{P_2}\right)^{n-1/n}$$

$$\frac{300}{T_2} = \left(\frac{110}{660}\right)^{1.3-1/1.3}$$

$$T_2 = 452.99\,K$$

Universal gas constant $= 8.314\,kJ/kgK$

$$R = \frac{\text{Universal gas constant}}{\text{Molecular weight}}$$

$$R = \frac{R_u}{M_w} = \frac{8.314}{30}$$

$$R = 0.277 \text{ kJ/kg K}$$

$$W = \frac{1 \times 0.277 (300 - 452.99)}{1.3 - 1}$$

$$\boxed{W = -141.26 \text{ KJ}}$$

Heat transfer $Q = W \times \left[\frac{\gamma - n}{\gamma - 1}\right]$

$$\gamma = C_P/C_V.$$

$$R = C_p - C_v$$

$$C_v = C_p - R$$

$$= 2.1 - 0.277$$

$$C_v = 1.82 \text{ KJ/kgK}$$

$$\gamma = C_P/C_V = 2.1/1.82$$

$$\gamma = 1.16$$

$$Q = -141.26 \left[\frac{1.16 - 1.3}{1.16 - 1}\right]$$

$$\boxed{Q = 135.72 \text{ KJ}}$$

b) Argon gas.

$$R = \frac{R_u}{M_w} = \frac{8.314}{40}$$

$$R = 0.207 \text{ kJ/kgK}$$

$$W = \frac{mR(T_1 - T_2)}{n - 1}$$

$$W = \frac{1 \times 0.207\,(300 - 452.99)}{1.3 - 1}$$

$$\boxed{W = -105.563 \text{ KJ}}$$

$$R = C_p - C_v$$

$$0.207 = 0.52 - C_v$$

$$C_v = 0.313$$

$$\gamma = C_p/C_v = \frac{0.52}{0.313}$$

$$\gamma = 1.661$$

$$Q = W\left[\frac{\gamma - n}{\gamma - 1}\right]$$

$$= -105.56\left[\frac{1.661 - 1.3}{1.661 - 1}\right]$$

$$\boxed{Q = -57.65 \text{ KJ}}$$

7. **A mass of 0.8 kg air at 1 bar 25°C it contained in a gas tight frictionless piston in cylinder device. The air is now compressed to a final pressure of 5 bar. During the process heat transferred from the air such that the temperature inside the cylinder remains constant. Calculate the heat transfer and work done during the process and direction of each in the process. (JAN 2015)**

given data

$m = 0.8\ kg$

$P_1 = 1\ bar = 100\ kN/m^2$

$T_1 = 25°c = 298\ K$

$P_2 = 5\ bar = 500\ kN/m^2$

Isothermally process $PV = C$

To find

Q, W and its direction.

Sol

characteristic gas equation

$P_1V_1 = mRT_1$ $\quad \therefore R = 0.287\ \frac{kJ}{kgK}$

$100 \times V_1 = 0.8 \times 0.287 \times 298$

$V_1 = 0.684\ m^3$

Pressure & volume relation for constant temperature

$\frac{P_1}{P_2} = \frac{V_2}{V_1}$

$\frac{100}{500} = \frac{V_2}{0.684}$

$V_2 = 0.136\ m^3$

Workdone $W = P_1V_1 \ln(V_2/V_1)$

$= 100 \times 0.684 \ln\left(\frac{0.136}{0.684}\right)$

$\boxed{W = -110.48\ KJ}$ Workdone on the system

$Q = W$ for Constant temperature Pressure process

$\boxed{Q = -110.48\ kJ}$ heat rejected from the system.

8. **A quantity of air having a volume of 0.04 $m^3$ at a temperature of 525 K and a pressure of 150 $N/cm^2$ is expanded at constant pressure to 0.08 $m^3$it is then expanded adiabatically to 0.012 $m^3$.**

**Find:** **a) Temperature and pressure at the end of the adiabatic process and**

**b) Work done during each stage(NOV 2017)**

given data:-

$V_1 = 0.04\ m^3$

$T_1 = 525K$

$P_1 = 150\ N/cm^2 \rightarrow 1500\ kn/m^2$

$V_2 = 0.08\ m^3$

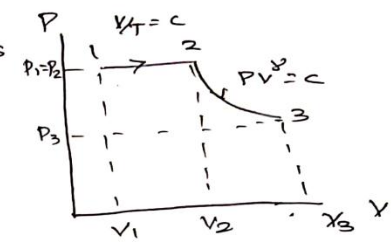
$V_3 = 0.012\ m^3$

Find:- a) Temp & pressure at the End of Adiabatic Expansion $T_3$, $P_3$

b) $W_{1-2}$ & $W_{2-3}$

Sol.

1-2 Constant pressure process




T & V relation

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

$$\frac{0.04}{0.08} = \frac{525}{T_2}$$

$$T_2 = 1050K$$

2-3 process. R.A.E.P T & V relation

$$\frac{T_3}{T_2} = \left(\frac{V_2}{V_3}\right)^{\gamma-1}$$

$$T_3 = T_2\ (V_2/V_3)^{\gamma-1} = T_3 = 1050 \times \left(\frac{0.08}{0.12}\right)^{1.4-1}$$

$$\boxed{T_3 = 892.79K}$$

P & V relation

$$\frac{P_3}{P_2} = \left(\frac{V_2}{V_3}\right)^{\gamma} = P_3 = P_2 \times (V_2/V_3)^{\gamma}$$

$$P_3 = 1500 \times (0.08/0.12)^{1.4}$$

$$\boxed{P_3 = 850.283\ kn/m^2.}$$

W = ?

$W_{1-2}$ is Constant pressure process.

1-2 Constant Pressure

$$W_{1-2} = P_1 (V_2 - V_1)$$

$$= 1500 (0.08 - 0.04)$$

$$W_{1-2} = 60 \text{ kJ}$$

2-3 = Reversible Adiabatic Process

$$W_{2-3} = \frac{P_1 V_1 - P_3 V_3}{\gamma - 1}$$

$$= \frac{1500 \times 0.04 - 850.283 \times 0.12}{1.4 - 1}$$

$$\boxed{W_{2-3} = 44.915 \text{ kJ}}$$

9. **A domestic refrigerator is loaded with food and the door closed. During a certain period the machine consumes 1 kWh of energy and the internal energy of the system drops by 5000 kJ. Find the net heat transfer for the system.(NOV 2018)**

given data

$$W = -1 \text{ kWh}$$

$$= -1 \times 3600$$

$$W = -3600 \text{ kJ}$$

$$\Delta U = U_2 - U_1 = -5000 \text{ kJ}$$

To find

$$Q = ?$$

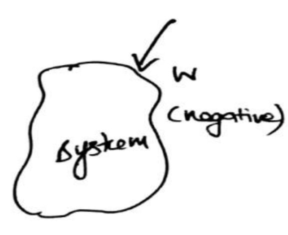
Sol

By first law of T.D

$$Q = W + \Delta U$$

$$= -3600 - 5000$$

$$\boxed{Q = -8600 \text{ kJ}}$$



10. **A mixture of gases expands at constant pressure from 1 MPa, 0.03 m$^3$ to 0.06 m$^3$ with 84 kJ positive heat transfer. There is no work other than that done on a piston. Find ΔE for the gaseous mixture. The same mixture expands through the same state path while a stirring device does 21 kJ of work on the system. FindΔE, W, and Q for the process. (NOV 2018)**

given data

$P_1 = 1\ MPa = 1000\ kPa$

$V_1 = 0.03 m^3$

$V_2 = 0.06 m^3$

$Q = 84\ KJ$

$W = -21\ KJ$

To find

$\Delta U = ?$

$\Delta U, W, Q$ for same mixture Expands the same state path while a stirring device.

Sol

$$\text{Workdone } (W) = \int_{V_1}^{V_2} P dV$$

$$= \int_{V_1}^{V_2} 1000 \times dV$$

$$= 1000 \int_{V_1}^{V_2} dV$$

$$= 1000 (V_2 - V_1)_{0.03}^{0.06}$$

$$W = 1000 (0.06 - 0.03)\ KJ$$

$$W = 30\ KJ$$

$Q = W + \Delta U$

$84 = 30 + \Delta U$

$\Delta U = 84 - 30\ KJ$

$\boxed{\Delta U = 54\ KJ}$

$W = -21\ KJ, \Delta U = 54\ KJ$ (for same mixture with same state)

$Q = W + \Delta U$

$Q = -21 + 54$

$\boxed{Q = 33\ KJ}$ $\boxed{W = -21\ KJ}$

11. **A steam turbine produces 500 MW of power. Steam enters the turbine at 12.5 MPa and 450°C. It leaves the turbine at 15 KPa and 0.91 quality. What is the mass flow rate of steam through the turbine if the heat loss to the surroundings from the turbine surface could be assumed as 5 MW? Show the process on *P-h* diagram.(MAY 2017)(Out of syllabus)**

12. **0.5 kg of air at a pressure of 1 bar occupies a volume of 0.4 m$^3$. If this air expands isothermally to a volume of 0.8 m$^3$ Find**
    **a) Initial temperature**
    **b) External work done**
    **c) Change in internal energy      Assume R = 0.29kJ/kgK(MAY 2018)**

Given data

$m = 0.5$ kg      $R = 0.29$ kJ/kgK.

$P_1 = 1$ bar $= 100$ kN/m$^2$

$V_1 = 0.4$ m$^3$

$V_2 = 0.8$ m$^3$

Isothermally process (or) constant temperature process $PV = C$

To find

$T_1$, $W$, $\Delta U$

Sol

a) Initial temperature $T_1$

from characteristic gas equation

$P_1V_1 = mRT_1$

$100 \times 0.4 = 0.5 \times 0.29 \times T_1$

$\boxed{T_1 = 275.86\text{K}}$

b) Workdone $W = P_1V_1 \ln(V_2/V_1)$

$= 100 \times 0.4 \times \ln(0.8/0.4)$

$\boxed{W = 27.72 \text{ KJ}}$

c) Change in internal Energy $(\Delta U)$

$\Delta U = mc_v(T_2 - T_1)$

$T_2 = ?$

from constant temperature process

$T_1 = T_2$.

$\Delta U = mc_v(T_2 - T_1)$

$= mc_v(T_1 - T_1)$

$\boxed{\Delta U = 0}$

13. **0.2 $m^3$ of air at 4 bar and 130°C is contained in a system. A reversible adiabatic expansion takes place till the pressure falls to 1.02 bar. The gas is then heated at constant pressure till enthalpy increases by 72.5 kJ. Calculate :**

**a) The work done**

**b) The index of expansion, if - the above processes are replaced by a single reversiblepolytropic process giving the same work between the same initial and final states.Takecp = 1 kJ/kg K, cv = 0.714 kJ/kg K.(NOV 2016)**

Given data

$V_1 = 0.2 m^3$

$P_1 = 4 bar = 400 kN/m^2$

$T_1 = 130°c = 403K$

$P_2 = 1.02 bar = 102 kN/m^2$

$\Delta H = 72.5 kJ$ $\quad c_p = 1 kJ/kg K$

$c_v = 0.714 kJ/kg K.$

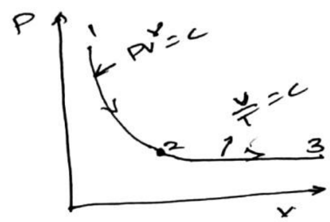
To Find

a) Workdone

b) n if the reversible polytropic process. giving the same work b/w same state.

Sol.

1-2 Reversible Adiabatic process.




Pressure & Volume relation

$$\frac{P_1}{P_2} = \left(\frac{V_2}{V_1}\right)^\gamma$$

$\gamma = cp/cv$

$= 1/0.714$

$\gamma = 1.4$

$$\frac{400}{102} = \left(\frac{V_2}{0.2}\right)^{1.4}$$

$$\frac{V_2}{0.2} = \left(\frac{400}{102}\right)^{1/1.4}$$

$V_2 = 0.53 m^3$

Also

$$\left(\frac{P_1}{P_2}\right)^{\gamma-1/\gamma} = \frac{T_1}{T_2}.$$

$$T_2 = \frac{T_1}{(P_1/P_2)^{\gamma-1/\gamma}} = \frac{403}{\left(\frac{400}{102}\right)^{1.4-1/1.4}}$$

$T_2 = 272.7 K$

$$W_{1-2} = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

$$= \frac{400 \times 0.2 - 102 \times 0.53}{1.4 - 1}$$

$$W_{1-2} = 64.85 \text{ KJ}$$

2-3 Process Constant pressure.

$$\Delta H_{2-3} = m\, c_p (T_3 - T_2)$$

$$m = ?$$

$$R = c_p - c_v$$
$$= 1 - 0.714$$
$$= 0.286 \text{ KJ/kg K}$$

$$P_1 V_1 = m R T_1$$

$$400 \times 0.2 = m \times 0.286 \times 403$$

$$\underline{\underline{m = 0.694 \text{ kg}}}$$

$$\Delta H_{2-3} = m\, c_p (T_3 - T_2)$$

$$72.5 = 0.694 \times 1 (T_3 - 272.7)$$

$$\underline{\underline{T_3 = 377 \text{ K}}}$$

Volume & Temperature relation.

$$\frac{V_2}{V_3} = \frac{T_2}{T_3}$$

$$\frac{0.53}{V_3} = \frac{272.7}{377}$$

$$\underline{\underline{V_3 = 0.732}}\ m^3$$

Workdone

$$W_{2-3} = P_2 (V_3 - V_2)$$

$$= 102 \times (0.732 - 0.53)$$

$$W_{2-3} = 20.604 \text{ KJ}$$

Workdone by the path 1-2-3 given by

$$W_{1-2-3} = W_{1-2} + W_{2-3}$$

$$= 64.85 + 20.604$$

$$\boxed{W_{1-2-3} = 85.49 \text{ kJ}}$$

b) Index of Expansion (n) = ?
polytropic process

$$W_{1-2-3} = W_{1-3}$$

$$W_{1-3} = \frac{P_1V_1 - P_3V_3}{n-1}$$

$$85.49 = \frac{400 \times 0.2 - 102 \times 0.732}{n-1}$$

$$\boxed{n = 1.06}$$

14. **One kg of gas expands at constant pressure from 0.090 $m^3$ to 0.15 $m^{3.}$ If the initial temperature of the gas is 500 K, find the final temperature, net heat transfer, change in internal energy and the pressure of the gas. Assume $C_p$ = 1.005 kJ/kg K, $C_v$ = 0.71 kJ/kg K, R = 0.295 kJ/kg K. (SEP 2020)**

given data

$m = 1\ kg$

$V_1 = 0.090\ m^3$

$V_2 = 0.15\ m^3$

$T_1 = 500\ K$

$C_p = 1.005\ kJ/kg\ K$

$C_v = 0.71\ kJ/kg\ K$

$R = 0.295\ kJ/kg\ K$

Constant pressure process $\frac{V}{T} = C$

To find

$T_2,\ Q,\ \Delta U,\ P$

Sol

by characteristic gas equation

$P_1 V_1 = mRT_1$

$P_1 = \frac{mRT_1}{V_1}$

$P_1 = \frac{1 \times 0.295 \times 500}{0.090}$

$\boxed{P_1 = 1638\ kPa}$

$T_2 = ?$

Temperature volume relation

$\frac{V_1}{V_2} = \frac{T_1}{T_2}$.

$\frac{0.090}{0.15} = \frac{500}{T_2}$

$\boxed{T_2 = 833.33\ K.}$

$Q = mc_p(T_2 - T_1)$

$= 1 \times 1.005\ (833.33 - 500)$

$\boxed{Q = 334.996\ kJ}$

$\Delta U = mc_v(T_2 - T_1)$

$= 1 \times 0.71\ (833.33 - 500)$

$\boxed{\Delta U = 236.643\ kJ}$

15. **An air compressor requires a shaft work is 200 kJ/kg of air and the compression of are causes increases the enthalpy of air by 100 kJ/kg of an. Cooling water required for cooling the compressor picks up heat by 90 kJ/kg of air. Determine the heat transferred from compressor to atmosphere. (NOV 2016)**

Given data

Increase Enthalpy $h_2 = h_1 + 100$ kJ/kg

work = $-200$ kJ/kg

$Q = -90$ kJ/kg

Tofind

$Q_s = ?$ surrounding heat transfer

Sol

SFEE for Compressor

$$h_1 + (Q)_t = h_2 - W$$

$$\cancel{h_1} - (90 + Q_s) = \cancel{h_1} + 100 - 200$$

$$-90 - Q_s = 100 - 200$$

$$-Q_s = -100 + 90$$

$$-Q_s = -10 \text{ kJ/kg}$$

$$\boxed{Q_s = 10 \text{ kJ/kg}}$$

Surrounding heat transferred is 10 kJ/kg.

16. **Derive the expression of work done during isentropic process. (MAY 2018)**

Workdone for Isentropic process (or) adiabatic process $(PV^{\gamma} = C)$

$$W = \int_1^2 P\,dV \qquad PV^{\gamma} = C$$

$$W = \int_1^2 \frac{C}{V^{\gamma}}\,dV \qquad P = \frac{C}{V^{\gamma}}$$

$$= C\int_1^2 \frac{1}{V^{\gamma}}\,dV$$

$$= C\int_1^2 V^{-\gamma}\,dV$$

$$= C\left[\frac{V^{-\gamma+1}}{-\gamma+1}\right]_{V_1}^{V_2} \qquad PV^{\gamma} = C$$

$$P_1V_1^{\gamma} = P_2V_2^{\gamma} = C$$

$$= C\left[\frac{V_2^{-\gamma+1} - V_1^{-\gamma+1}}{-\gamma+1}\right]$$

$$= P_1V_1^{\gamma}\left[\frac{V_1^{-\gamma+1} - V_2^{-\gamma+1}}{\gamma-1}\right]$$

$$= \frac{P_1 \cancel{V_1^{\gamma}}\, V_1^{\cancel{-\gamma}+1} - P_2 V_2^{\cancel{-\gamma}+1} \times \cancel{V_2^{\gamma}}}{\gamma - 1}$$

$$\boxed{W = \frac{P_1V_1 - P_2V_2}{\gamma - 1}\ \text{kJ}}$$

or

$$PV = mRT$$

$$\boxed{W = \frac{mRT_1 - mRT_2}{\gamma - 1}\ \text{kJ}}$$

17. **In a compressor the air enter at 27°C at1 atm. and leaves at 227°C and 1 MPa. Determine the work done per unit mass of air assuming the velocities at entry and exit to be negligible. Also determine the additional work required if velocities are 10 m/s and 50 m/s, at inlet and outlet respectively. (MAY 2019)**

given data

I

$T_1 = 27^\circ c = 300 K$

$T_2 = 227^\circ c = 500 K.$

$P_1 = 1 MPa$

II

$c_1 = 10 m/s$

$c_2 = 50 m/s$

To find

work done per kg

Addition work done if considered velocities

Sol

From steady flow Energy Equation

$\frac{c_1^2}{2000} + P_1V_1 + U_1 = \frac{c_2^2}{2000} + P_2V_2 + U_2 + W.$

Velocity are negligible. $C_1 = C_2 = 0.$

$P_1 V_1 + U_1 = \cancel{c_2} \; P_2 V_2 + U_2 + W.$

$h_1 = P_1 v_1 + U \quad ; \quad h_2 = P_2 V_2 + U_2.$

$h_1 = h_2 + W.$

$W = h_1 - h_2.$

$h_1 = CpT_1 \quad ; \quad h_2 = CpT_2.$

$W = CpT_1 - CpT_2$

$= Cp(T_1 - T_2)$

$= 1.005 (300 - 500)$

$\boxed{W = -201 \text{ KJ/kg.}}$

if considered velocities.

Steady flow Energy Equations.

$$\frac{C_1^2}{2000} + h_1 = \frac{C_2^2}{2000} + h + W$$

$$\frac{10^2}{2000} + CpT_1 = \frac{50^2}{2000} + CpT_2 + W.$$

$$W = Cp(T_1 - T_2) + \frac{10^2 - 50^2}{2000}$$

$$= 1.005(300 - 500) + \frac{10^2 - 50^2}{2000}$$

$$= -201 - 1.2$$

$$\boxed{W = -202.2 \text{ KJ/kg}}$$

18. **At the inlet to a certain nozzle the enthalpy of fluid passing is 2800 kJ/kg and the velocity is 50 m/s. At the discharge end the enthalpy is 2600 kJ/kg. The nozzle is horizontal and there is negligible heat loss from it.**

   a. **Find the velocity at exit of the nozzle.**

   b. **If the inlet is 900 cm$^2$ and the specific volume at inlet is 0.187 m$^3$/kg, find the mass flow rate**

   c. **If the specific volume at the nozzle exit is 0.498m$^3$ /kg, find the exit area of nozzle (NOV 2016)**

**Given data**

$h_1 = 2800$ KJ/Kg

$C_1 = 50$ m/s

$h_2 = 2600$ KJ/Kg

$A_1 = 900 \text{ cm}^2 = 0.09 \text{ m}^2$

$v_1 = 0.187 \text{ m}^3/\text{kg}$

$v_2 = 0.498 \text{ m}^3/\text{kg}$

**To find** $C_2,\ m,\ A_2 = ?$

**Sol**

SFEE for nozzle

$$h_1 + \frac{C_1^2}{2000} = h_2 + \frac{C_2^2}{2000}$$

$$h_1 - h_2 = \frac{C_2^2 - C_1^2}{2000}$$

$$2800 - 2600 = \frac{C_2^2}{2000} - \frac{50^2}{2000}$$

$$\frac{C_2^2}{2000} = 200 + 1.25$$

$$\boxed{C_2 = 634.42 \text{ m/s}}$$

$$m = \frac{A_1 C_1}{v_1} = \frac{0.09 \times 50}{0.187}$$

$$\boxed{m = 24.06 \text{ kg}}$$

$$m_1 = m_2$$

$$\frac{A_1 C_1}{v_1} = \frac{A_2 C_2}{v_2}$$

$$\frac{0.09 \times 50}{0.187} = \frac{A_2 \times 634.42}{0.498} = 24.06 = 1269.91 A_2$$

$$\boxed{A_2 = 0.0189 \text{ m}^2}$$

19. **A turbine operates under steady flow conditions receiving steam at the following state. Pressure 1.2 Mpa, temperature 188$^o$C, enthalpy 2785 kJ/kg, velocity 33.3m/s and elevation 3m. The steam leaves the turbine at the following state. Pressure 20kPa, enthalpy 2512kJ/kg, velocity 100 m/s and elevation 0m. Heat flow to the surrounding is at the rate of 0.29kJ/s. If the rate of steam flow through the turbine is 0.42 kg/s. Determine the power output of the turbine in kW. (SEP 2020)**

Given data

$P_1 = 1.2\ \text{MPa} = 1200\ \text{kPa}$

$T_1 = 188^\circ C = 461\ K$

$h_1 = 2785\ \text{KJ/kg}$

$C_1 = 33.3\ \text{m/s}$

$Z_1 = 3\ \text{m}$

$P_2 = 20\ \text{kPa}$

$h_2 = 2512\ \text{KJ/kg}$

$C_2 = 100\ \text{m/s}$

$Z_2 = 0$

$Q = -0.29\ \text{KJ/s}$

$m = 0.42\ \text{kg/s}$

To Find

$P = ?$

Sol

Steady flow Energy Equation

$$m\left(\frac{gz_1}{1000} + \frac{1}{2000}c_1^2 + h_1 + Q\right) = m\left(\frac{gz_2}{1000} + \frac{1}{2000}c_2^2 + h_2 + W\right)$$

$$0.42\left[\frac{9.81\times 3}{1000} + \frac{1\times 33.3^2}{2000} + 2785 - 0.29\right] = 0.42\left[\frac{1\times 100^2}{2000} + 2512 + W\right]$$

$$W = 0.42\left[\frac{9.81\times 3}{1000} + \frac{33.3^2 - 100^2}{2000} + 2785 - 2512 - 0.29\right]$$

$W = 265.29$

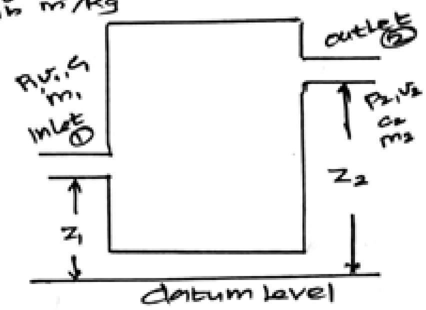
$P = 265.29 \times 0.42$

$$\boxed{P = 111.42\ \text{KW}}$$

20. **Derive the steady flow energy equation and reduce it for turbine, compressor, nozzle and boiler. (MAY 2016)**

Consider an open system through which the working substance flows as steady rate the working substance ntering the system at 1 and leaves the system at 2

$P_1 \rightarrow$ Pressure at inlet in bar
$v_1 \rightarrow$ specific Volume at inlet in $m^3/kg$
$c_1 \rightarrow$ Velocity at inlet $m/s$
$u_1 \rightarrow$ Internal Energy at Inlet in KJ/kg
$Z_1 \rightarrow$ height (or) datum above the level for inlet in m
$m_1 \rightarrow$ mass at inlet in kg




$P_2, v_2, c_2, u_2, z_1$ Corresponding Value for working substance leaving the system.
$Q \rightarrow$ heat supplied to the system in KJ/kg
$W \rightarrow$ work delivered by the system in KJ/kg

total Energy Entering the system =
Potential Energy + kinetic Energy + flow Energy + Internal Energy + heat Energy

$$= gz_1 + \frac{c_1^2}{2} + u_1 + P_1 V_1 + Q$$

total Energy leaving the System =
Potential Energy + kinetic Energy + flow Energy + Internal Energy + work.

$$gz_2 + \frac{c_2^2}{2} + u_2 + P_2 V_2 + W.$$

total Energy Entering the System = total Energy leaving the system.

$$gz_1 + \frac{c_1^2}{2} + P_1V_1 + U_1 + Q = gz_2 + \frac{c_2^2}{2} + P_2V_2 + U_2 + W.$$

$$h = U + PV, \quad h_1 = U_1 + P_1V, \quad h_2 = U_2 + P_2V_2$$

$$gz_1 + \frac{c_1^2}{2} + h_1 + Q = gz_2 + \frac{c_2^2}{2} + h_2 + W$$

the above Equation is known as steady Flow Energy Equation (SFEE) the above Equation represents the Energy Flow per unit of mass of the working Substance (J/Kg) when the Equation is multiplied by mass of the working substance throughout then all term will represents the Energy Flow per unit time (J/s)

Then above equation becomes

$$m\left(gz_1 + c_1^2/2 + h_1 + Q\right) = m\left(gz_2 + c_2^2/2 + h_2\right) + W.$$

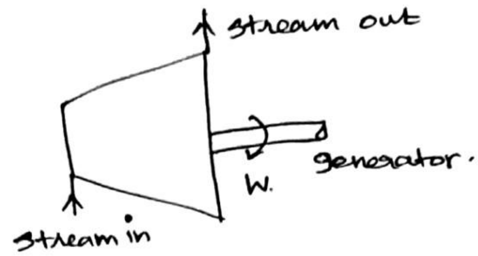
(or)

$$m\left(gz_1 + c_1^2/2 + P_1V_1 + U_1 + Q\right) = m\left(gz_2 + c_2^2/2 + P_2V_2 + U_2\right) + W$$

if the values of Q and W is KJ/kg and $h_1$ and $h_2$ are substituted in KJ/kg Then above equation becomes

$$m\left(\frac{gz_1}{1000} + \frac{c_1^2}{2000} + P_1V_1 + U_1\right) + Q = m\left(\frac{gz_2}{1000} + \frac{c_2^2}{2000} + P_2V_2 + U_2\right) + W.$$

## Steady flow Energy Equation for turbine




In a stream or gas turbine steam or gas is passed through the turbine and part of its Energy is converted into work this output of the turbine runs a generator to produce electricity

Applying energy equation to the turbine.

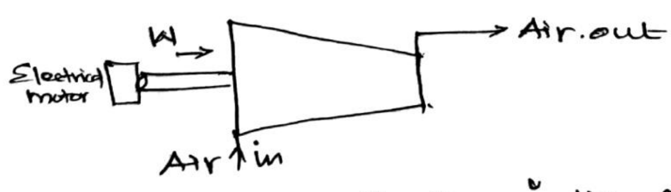
$$z_1 = z_2$$

$$h_1 + \frac{c_1^2}{2} - Q = h_2 + \frac{c_2^2}{2} + W.$$

the sign of Q is negative because heat is rejected

## Compressor

A compressor compresses air and supplies the same at moderate pressure and in large quantity.




Applying energy Equation to the system.

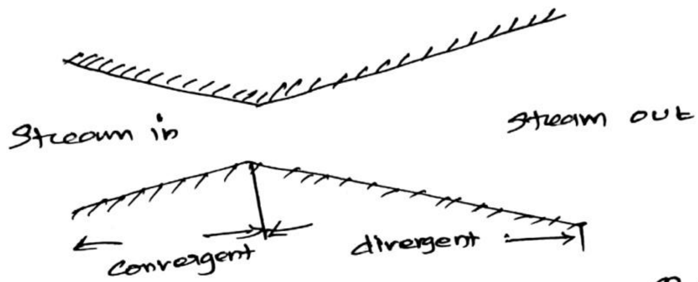
$$\Delta z = 0$$

$$\left[h_1 + \frac{c_1^2}{2}\right] - Q = \left[h_2 + \frac{c_2^2}{2}\right] - W.$$

the Q is taken as negative as heat is lost from the system.

Steam nozzle

In case of a nozzle as the Enthalpy of the fluid decrease and pressure drops Simultaneously the flow of fluid is accelerated this is generally used to convert the part of the Energy of steam into kinetic Energy of steam supplied to the turbine.




potential Energy = 0  W = 0  Q = 0.

Applying the Energy Equation to the system

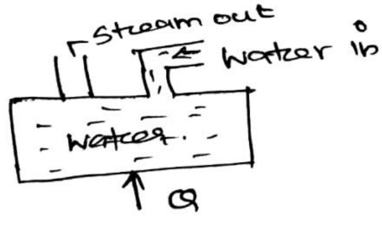
$$h_1 + \frac{c_1^2}{2} = h_2 + \frac{c_2^2}{2}.$$

$$c_2 = \sqrt{c_1^2 + 2(h_1 - h_2)}$$

$c_1$ is less then $c_2$  $(c_1 << c_2)$

$$c_2 = \sqrt{2(h_1 - h_2)}$$

Boiler




$\Delta z = 0$  kinetic Energy is zero

$W = 0$ since neigther any work is developed nor absorbed.

A. boiler transfers heat to the incoming ~~air~~ ~~and~~ water and generates the steam.

Applying Energy Equation to the system

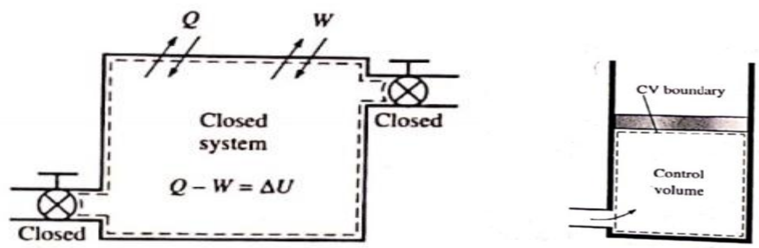
$h_1 + Q = h_2$  $Q = h_2 - h_1$.

21. **Explain a unsteady flow process.(MAY 2019)**

During a steady flow process, no changes occur within the Control Volume thus one does not need to be concerned about what is going on within the boundaries. Not having to worry about any changes within the Control Volume with time greatly Simplifities the analysis.

Many processes of interest, however, involve Change within the Control volume with time. Such processes are called unsteady flow, or transient flow processes. the steady flow relations developed earlier are obviously not applicable to these processes. When an unsteady flow process is analyzed, it is Important to keep track of the mass and Energy Contents of the Control volume as well as the Energy Interaction across the boundary.

unsteady - flow process start and end over some finite time period Instead of continuing Indefinitly. therefore in this section, we deal with changes (chang per unit time). An unsteady - flow system in some respects is similar to a closed system except that the mass within the system boundaries does not remain Constant during a process




Another difference between steady and unsteady flow system is that steady flow system are fixed in space, size, and shape unsteady flow systems however, are not. They are usually stationary that is they are fixed in space, but they may involve moving boundaries. and thus boundary work.

The mass balance for any system undergoing any process can be expressed.

$$m_{in} - m_{out} = \Delta m_{system}$$

Where $\Delta m_{system} = m_{final} - m_{initial}$ is the change in the mass of the system for control volumes, it can be expressed more explicity as.

$$m_i - m_e = (m_2 - m_1)_{c.v}$$

Where $i$ = inlet, $e$ = exit $1$ = initial state and $2$ = final state of the control volume. Often one or more terms in the equation above are zero for example $m_i = 0$ if no mass enters the control volume during the process $m_e = 0$ if no mass leaves and $m_1 = 0$ if the control volume is initially evacuated.

The Energy content of a control volume changes with time during an unsteady flow process. The magnetitude of change depends on the amount of Energy transfer across system boundaries as heat and work as well as on amount of Energy tranported into and out of the Control Volume by mass during the process. When analyzing an unsteady - flow process, We must Keep track of the Energy content of the Control Volume as well as the Energies of the Incoming and outgoing flow streams.

the general Energy balance was given earlier as

$$\underbrace{E_{in} - E_{out}}_{\text{net Energy transfer by heat, work and mass.}} = \underbrace{\Delta E_{system}}_{\text{change in internal kinetic, potential, etc. Energies}} \quad (kJ)$$

The general unsteady - flow process. In general, is difficult to analyse because the properties of the mass at the inlets and exists may change during a process. Most unsteady flow process however, can be represented reasonably well by the uniform flow process. which Involves the following Idealization. the fluid flow at any inlet or exist is uniform and steady and thus The fluid properties do not change with time or position over the cross section of an Inlet or exit if they do they are averaged and treated as constants for the Entire process.

22. **A vessel contains superheated steam at 6 MPa and 500°C. The vessel is cooled until the pressure of steam drops to 10 bar. Determine the entropy change and heat transfer for the process per unit mass of steam (MAY 2017)**

given data

$P_1 = 6 MPa = 6000 kPa$

$T_1 = 500^\circ c = 773 K$

$P_2 = 10 bar = 1000 kPa$

To find

$\Delta S, Q$

<u>sol</u>

Constant Volume Process

temperature pressure

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

$$\frac{6000}{1000} = \frac{773}{T_2}$$

$$T_2 = 128.83 K.$$

$$\Delta S = m c_v \ln (P_2/P_1)$$

$$= 1 \times 0.71 \ln (6000/1000)$$

$$\boxed{\Delta S = 1.272 \ kJ/kgK}$$

$$Q = m c_v (T_2 - T_1)$$

$$= 1 \times 0.71 (128.83 - 773)$$

$$\boxed{Q = -457.3607 \ kJ/kg}$$